A Dual-Discriminator Generative Adversarial Network for Anomaly Detection

Da Ding, Youquan Wang, Haicheng Tao, *Member, IEEE*, Jia Wu, *Senior Member, IEEE*, and Jie Cao

***Abstract*—Multivariate time series anomaly detection has shown potential in various fields, such as finance, aerospace, and security. The fuzzy definition of data anomalies, the complexity of data patterns, and the scarcity of abnormal data samples pose significant challenges to anomaly detection. Researchers have extensively employed autoencoders (AEs) and generative adversarial networks (GANs) in studying time series anomaly detection methods. However, relying on reconstruction error, the AE-based anomaly detection algorithm needs more effective regularization methods, rendering it susceptible to the problem of overfitting. Meanwhile, GAN-based anomaly detection algorithms require high-quality training data, significantly impacting their practical deployment. We propose a novel GAN based on a dual-discriminator structure to address these issues. The model first processes the data with the generator to obtain the reconstruction error and then calculates pseudo-labels to divide the data into two categories. One data category is input into the first discriminator, where a minor loss between the data and its reconstructed counterpart is better. The other data category is input into the second discriminator, where a larger loss between the data and its reconstructed counterpart is better. Through this process, the model can effectively constrain the generator, retaining information on normal data during data reconstruction while discarding information on abnormal data. After conducting experiments on multiple benchmark datasets, the proposed GAN based on a dual-discriminator structure achieved good results in anomaly detection, outperforming several advanced methods. Additionally, the model also performed well in practical transformer data.**

***Index Terms*—Anomaly detection, dual-discriminator structure, generative adversarial networks (GANs), industrial applications, pseudo-labels.**

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I. INTRODUCTION

IN RECENT years, the proliferation of the Internet of Things (IoT) technology has driven the development of smart grids [1], [2] and intelligent transportation systems [3], [4]. As shown in Fig. 1, a large amount of system operation data is collected by deploying sensors and actuators. Accurately and timely identifying abnormal information within this data is crucial for taking corrective measures, addressing issues, and minimizing economic losses and human casualties [5]. Therefore, anomaly detection has emerged as a critical research area, aiming to precisely and promptly locate anomalous patterns within the collected information. As depicted in Fig. 2, an anomaly is a deviation from a pattern recognized by human cognition, resulting in a perceptual change (e.g., appearance corruption, style change, blurring, or occlusion) or a semantic change (e.g., class change or inversion behavior) to the pattern. Anomalies in time series are usually divided into point anomalies and collective anomalies; as illustrated in Fig. 2(a), a point anomaly is a type of anomaly when the data of a single point exceed the range of normal data distribution. Collective anomaly, as shown in Fig. 2(b), is a set of continuous data point sequences as a whole that is considered anomalous even though individual data points may not be anomalous. Threshold-based judgment is the most straightforward approach to anomaly detection, which proves feasible when applied to individual time series. However, as the number of collection devices and detection targets increases, along with the exponential growth in collected data volume and dimensionality, the interdependencies among the data become more intricate. Consequently, the threshold-based method becomes ineffective, necessitating the development of novel approaches to address this challenge. Hence, this study proposes an effective method for anomaly detection in multivariate time series data to address the aforementioned challenges.

IoT has provided abundant data for the advancement of deep learning. Concurrently, deep learning has offered numerous methods for detecting anomalies in IoT data [6], [7], [8]. For instance, clustering algorithms treat data that deviate from cluster centers as anomalies, while classification algorithms identify data that do not conform to distribution patterns as anomalies. However, these methods either fail to find the correlations among time series data or require additional information about anomalous data, rendering them unsuitable for high-dimensional data and large-scale series anomaly detection. Moreover, to handle large amounts of data, supervised methods necessitate the laborious task of effectively labeling the data, requiring substantial resources. Consequently, unsupervised methods have garnered significant

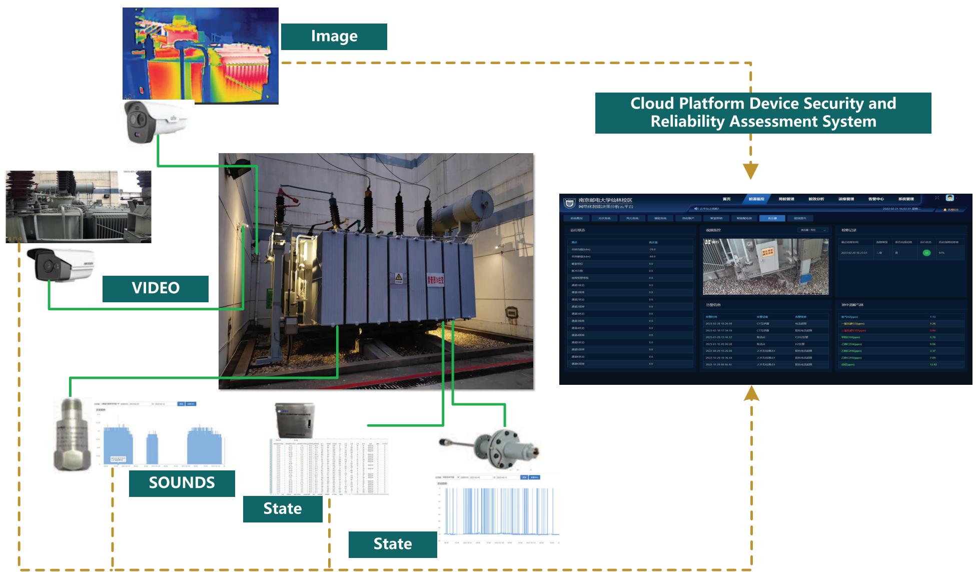

Fig. 1. Comprehensive transformer anomaly detection system.

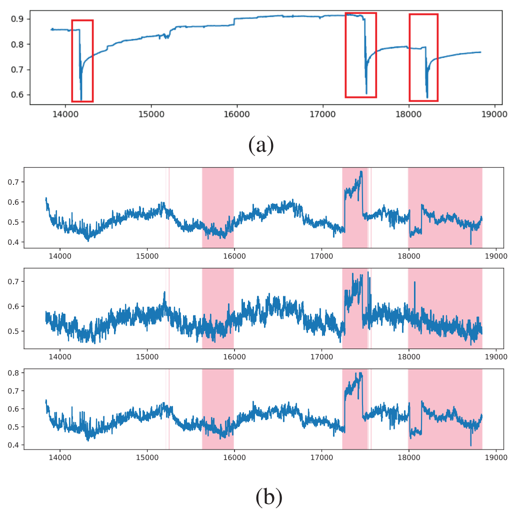

Fig. 2. Difference between univariate time series and multivariate time series in anomaly detection. (a) Univariate time series with point anomaly. (b) Multivariate time series with collective anomalies.

attention in the industry [9], [10], [11]. A commonly used approach in unsupervised anomaly detection is to map data into a low-dimensional space and then map the data in the low-dimensional space back to a high-dimensional space. This process allows normal data to retain information during the reconstruction phase while causing anomalous data to lose information. By employing appropriate reconstruction techniques, precise data reconstruction can be achieved, resulting in significant reconstruction errors for anomalous data. This approach enables cost-effective and highly accurate anomaly detection [12].

Anomaly detection involves the handling of nonlinear data with complex temporal correlations; thus, deep learning methods are used in this field. Generative adversarial networks (GANs) have emerged as promising techniques for unsupervised anomaly detection. The pioneering work of Goodfellow et al. [13] introduced GANs, which simultaneously train a generator and a discriminator. The generator aims to encode synthetic data based on the original data, while the discriminator distinguishes between samples of generated data and samples of original data. Recent work in anomaly detection has also explored the use of adversarial training [11], [14], [15], [16]. The AnoGAN method, proposed by Schlegl et al. [14], addresses the challenge of unsupervised anomaly detection. It utilizes deep convolutional learning to capture the characteristics of normal images and introduces a novel scoring scheme for mapping them to the latent space. However, this approach has high computational complexity. Zenati et al. [17] developed a bidirectional GAN (BiGAN) model, which showed improved performance in mapping images to the latent space compared to prior work. We propose a novel GAN architecture called TadGAN [18] that uses an autoencoder (AE) with an $L2$ objective function to reconstruct time series and evaluate errors to identify anomalies. TadGAN is trained on clean training sets, i.e., sets containing only normal samples. To address the impact of polluted data, the author proposes FGANmoraly [10], which filters out possible abnormal sample discriminators with pseudo-labels before training. However, this approach may lead to false positives and result in wastage of resources. Motivated by the aforementioned research contributions, this article presents a more comprehensive approach that effectively learns both the generator and the discriminator while making optimal use of the available data and minimizing the impact of contaminated data on the model.

This article proposes the novel dual-discriminator GAN (DDGAN) for anomaly detection. The main goal is to train a generator that can accurately reconstruct data while enlarging the reconstruction error between abnormal data and reconstructed abnormal data. Our approach involves the following key components.

1) Given the intrinsic temporal characteristics and intricate correlations present in multitime series data, we employ

an LSTM-based AE to facilitate the reconstruction of such data.

2) Considering the specific data requirements of GAN networks, we propose a dual safeguard mechanism consisting of a filter and a dynamic dictionary, which aims to mitigate the potential introduction of contaminated data.
3) To fully exploit the available data, we design two discriminators: one that receives normal data labeled as 0 and another that receives anomalous data stored in the dynamic dictionary.
4) We propose a new training loss function that considers the constraints imposed by the two discriminators on the AE to achieve more accurate data reconstruction.

The remainder of the article is organized as follows. Section II presents an overview of related research in the field of anomaly detection. Section III presents the relevant definitions used in this article. Section IV provides a more detailed exposition of the DDGAN model. Section V explains how the model is used for anomaly detection and presents the experimental results. Section VI summarizes the principal findings of this article and proposes potential avenues for future research.

II. Related Work

In recent years, the increase in data types, changes in application scenarios, and diversification of anomaly types have posed greater demands on anomaly data detection. The simplest method for anomaly detection involves threshold-based judgments [19]. However, with the growing complexity of interdata correlations and the diversification of data types, this straightforward approach is no longer applicable to the current complex environment of anomaly data detection. Consequently, researchers have shifted their focus to the study of unsupervised anomaly detection methods. These methods can be categorized into one-class classification, nearest neighbor-based algorithms, and reconstruction-based methods [20]. This article provides a detailed investigation and research on these methods.

Currently, research on anomaly data detection methods based on one-class classification has been extensively conducted. One-class detection entails learning the discriminative boundary of normal data, considering any value deviating from the data boundary as an anomaly. Classical algorithms include one-class SVM [8], [21] and isolation forest [22]. Ruff et al. [23] introduced a deep support vector method based on deep learning and one-class classification, where the model is trained by optimizing the objectives of anomaly detection. Miao et al. [24] introduced distributed online OCSVM, a distributed online one-class classification method capable of detecting anomaly data in distributed data. However, the aforementioned methods exhibit subpar performance when applied to high-dimensional data.

Nearest neighbor-based algorithms identify objects as anomaly data by measuring the distances between different entities, considering objects that are far from other data points as anomalies. The nearest neighbor-based algorithms can be further categorized into distance- and density-based methods. Distance-based methods define the neighbors of an object using a given radius and determine the anomaly score based on the number of neighbors [25]. On the other hand, density-based methods detect anomaly data based on the density of neighbors surrounding an object. Mahela et al. [26] proposed a network anomaly detection method based on fuzzy clustering, while Xing et al. [27] presented K-means clustering and weighted K-nearest neighbor regression-based algorithm to detect anomaly footprints of transmission line. Parwez et al. [28] introduced K-means clustering and hierarchical clustering to analyze anomalous behavior of mobile wireless networks. However, the implementation of nearest neighbor-based methods requires prior knowledge regarding the duration and quantity of anomalies while also disregarding the temporal dimension of interrelationships among time series data.

The reconstruction-based methods involve mapping the original data to high-dimensional or low-dimensional space using a constructed model and then evaluating data anomalies based on the error between the reconstructed data and the original data. However, this method may need more information when mapping abnormal data to low-dimensional space, leading to inaccurate reconstruction and larger errors. Principal component analysis (PCA) is a classic algorithm of this type, but it is limited to linear data and requires a Gaussian distribution. Several methods such as AE [29], [30], and variational long short-term memory (VLSTM) [31] have been proposed to address the above limitation. These methods lack effective regularization and may suffer from overfitting, resulting in poor performance [32]. GAN introduces adversarial regularization to alleviate overfitting, but this approach assumes an uncontaminated training set, and it fails to effectively model the data distribution when a significant amount of contaminated data is present in the training set [13]. To avoid capturing the distribution of anomalous data, the model employs a filtering mechanism on the discriminator using pseudo-labels for potential anomalous samples before training. However, this approach can lead to data waste and the possibility of mislabeled instances due to the utilization of pseudo-labels, resulting in suboptimal performance.

III. Preliminaries

Definition 1 (Multivariate Time Series): The dataset $D = \{D_1, D_2, \ldots, D_N\}$ consists of N sequential data instances, each D_i having a fixed length of K, noted as $D_i = \{x_i^1, x_i^2, \ldots, x_i^K\}$. The time intervals between consecutive data points are uniform, ranging from seconds to hours.

Definition 2 (Reconstruction): Reconstruction involves mapping the original data D into a low-dimensional space and then mapping it back into a high-dimensional space. The original data are denoted as $D = \{D_1, D_2, \ldots, D_N\}$, $D_i = \{x_i^1, x_i^2, \ldots, x_i^K\}$, while the data representation after reconstruction is denoted as $\hat{D} = \{\hat{D}_1, \hat{D}_2, \ldots, \hat{D}_N\}$, $\hat{D}_i = \{\hat{x}_i^1, \hat{x}_i^2, \ldots, \hat{x}_i^K\}$.

Definition 3 (Reconstruction Error): The reconstruction error quantifies the distance between the input data D_N and the reconstructed data $\hat{D}_N$, and it will be mathematically expressed as

$$e_N = \left\| D_N - \hat{D}_N \right\|. \quad (1)$$

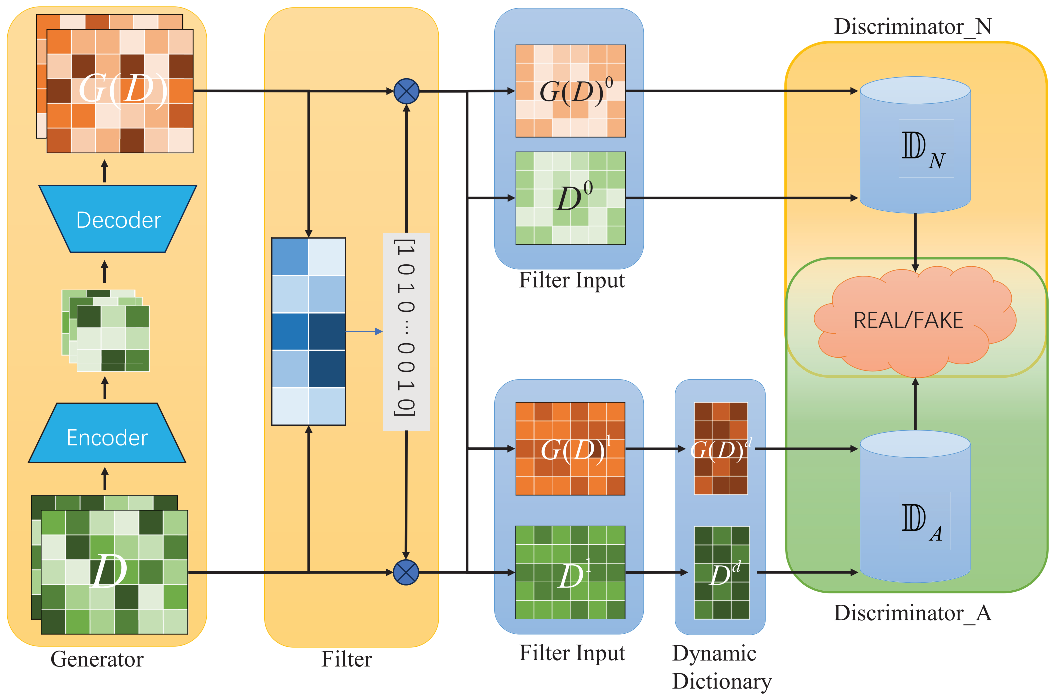

Fig. 3. Data D are passed through an encoder and decoder in the generator to obtain reconstructed data $G(D)$. Based on the reconstruction error E between D and the reconstructed data $G(D)$, the filter calculates pseudo-labels for the data. The data with a pseudo-label of 0 are fed into the discriminator $\mathbb{D}_N$, while the data with a pseudo-label of 1 undergo further updates in the dynamic dictionary and serve as input for the discriminator $\mathbb{D}_A$. The discriminators are subsequently trained using these refined samples.

Problem Statement: By performing a transformation on the original data, denoted as $D = \{D_1, D_2, \ldots, D_N\}$ and $D_i = \{x_i^1, x_i^2, \ldots, x_i^K\}$, where N represents the sampling time, anomaly detection assigns a label $y \in \{0, 1\}$ (0 for normal and 1 for abnormal) to each data point at a given sampling time. Reconstructing the inputted original data as $\hat{D}$=$\{\hat{D}_1, \hat{D}_2, \ldots, \hat{D}_N\}$ and $\hat{D}_i = \{\hat{x}_i^1, \hat{x}_i^2, \ldots, \hat{x}_i^K\}$. The anomaly score of the data is calculated by computing the reconstruction error between the input data D and the reconstructed data $\hat{D}$. Based on this anomaly score, a determination is made regarding whether the original data is anomalous or not.

IV. Methodology

In this section, more details on the implementation of the DDGAN will be presented.

A. Overview of the Proposed Model

This section will provide a detailed description of the DDGAN, which comprises two main parts: training and detection. The architecture of DDGAN is depicted in Fig. 3, and it consists of a generator, a filter, a dynamic dictionary, and two discriminators. $\mathcal{X}$ and $\mathcal{D}$ represent the feature space and latent vector space, respectively. The generator G first maps the observations into latent vector space and then reconstructs them into the feature space

$$\begin{aligned} f_{\text{en}} &: \mathcal{X} \to \mathcal{D} \\ f_{\text{de}} &: \mathcal{D} \to \mathcal{X}. \end{aligned} \tag{2}$$

Specifically, the encoder is an LSTM network, while the decoder is symmetric to the encoder.

On the other hand, the discriminators, denoted by $\mathbb{D}_A$ and $\mathbb{D}_N$, are trained based on the original data and reconstructed data, respectively. The original data are labeled as real, and the reconstructed data are labeled as fake to train discriminator $\mathbb{D}_A$ and discriminator $\mathbb{D}_N$. The discrimination mappings f_{dis_A} and f_{dis_D} are

$$\begin{aligned} f_{\text{dis}_A} &: \mathcal{X} \to [0, 1] \\ f_{\text{dis}_D} &: \mathcal{X} \to [0, 1]. \end{aligned} \tag{3}$$

The discriminator utilizes a ladder feed-forward network to accept time series data and produce results. Prior to inputting the data into the discriminator $\mathbb{D}_N$, a pseudo-label filter is utilized to remove significant abnormal data from the original and reconstructed data. Additionally, a dynamic dictionary is employed to store the most conspicuous portion of the filtered abnormal data, which is subsequently inputted into the discriminator $\mathbb{D}_A$. This process enables effective training of the generator under the dual constraints of discriminators $\mathbb{D}_A$ and $\mathbb{D}_N$.

B. Pseudo-Labels

GANs can learn the underlying distribution of training data and generate samples that follow that distribution. However, the exclusive training of GAN models on normal data results in limited diversity in generated samples, as the absence of exposure to abnormal samples during training restricts the model's ability to generate diverse samples. Similarly, training GANs on abnormal data only leads to the generation of abnormal samples while being ineffective in generating normal samples due to the lack of training on normal data. Furthermore, the inclusion of abnormal samples in the training set may introduce errors in the learned data distribution, leading to reduced accuracy in data detection.

A method involving the use of pseudo-labeling and dynamic arrays is proposed to filter and optimize the input to the discriminator. Pseudo-labeling allows discriminator $\mathbb{D}_N$ to receive "pseudo-normal data" as input, while dynamic arrays enable discriminator $\mathbb{D}_A$ to receive "pseudo-abnormal data" as input. By leveraging pseudo-labeling and dynamic arrays, the vision of training the discriminator on a single type of data is achieved, overcoming the challenges of obtaining normal or abnormal samples in practical applications.

Furthermore, a reconstruction method is adopted to calculate the distance between normal and abnormal data distributions. This method aims to minimize the error loss between normal data and reconstructed data while increasing the error loss between abnormal data and reconstructed data. This approach allows for the generation of abnormal samples while maintaining the fidelity of normal samples, thus addressing the limitations of GANs in generating diverse samples from both normal and abnormal data distributions. The specific process is presented in Algorithm 1. The original data $D = \{D_1, D_2, \ldots, D_N\}$, and $D_i = \{x_i^1, x_i^2, \ldots, x_i^K\}$ is input into G to obtain reconstructed data $\hat{D}=\{\hat{D}_1, \hat{D}_2, \ldots, \hat{D}_N\}$, $\hat{D}_i = \{\hat{x}_i^1, \hat{x}_i^2, \ldots, \hat{x}_i^K\}$. Based on Equation (1), the error $E = \{e_1, e_2, \ldots, e_N\}$ based on define between the original data D and the reconstructed data $G(D)$ can be expressed as

$$e_w = \left\| D_w - \hat{D}_w \right\|. \tag{4}$$

The mean and variance of E are represented by $M(E)$ and $V(E)$, respectively, enabling the calculation of the z-score z_w

$$z_w = \frac{(e_w - M(E))}{V(E)}. \tag{5}$$

Based on this z-score, the probability of x being anomalous is designed as

$$P(D_w) = \sigma(z_w). \tag{6}$$

As shown in Equation (6), if the error e_w is equal to the mean $M(E)$ or the z-score z_w equals 0, the probability of the data $P(D_w)$ being labeled as anomalous is 0.5. If the error e_w is greater than the mean $M(E)$, the probability of the data $P(D_w)$ being labeled as anomalous is greater than 0.5. Conversely, if the error e_w is smaller than the mean $M(E)$, the probability of the data $P(D_w)$ being labeled as anomalous is smaller than 0.5. This approach provides a robust method for detecting anomalous data.

During the initial training phase, the model may need to acquire more information to represent the latent information of the data accurately. Therefore, the same parameters are set for all data in the initial stage. In the subsequent phase, an impact factor θ is introduced to influence the anomaly probability; the factor θ should be consistent for all data under the initial conditions, showing more influence with the increase of training. Therefore, the factor θ is defined as

$$\theta = 1 - \frac{1}{S_{\text{epoch}}} \tag{7}$$

where S_{epoch} is the function about the current iteration and satisfies the following:

$$S_{\text{epoch}} = \begin{cases} 1, & \text{if } n = 1 \\ \infty, & \text{if } n \to \infty. \end{cases} \tag{8}$$

Combining Equation (6) and Equation (8), we can get the following:

$$\begin{aligned} P(x_w) &= \sigma(z_w * \theta) \\ &= \sigma\left(\frac{(e_w - M(E))}{V(E)} * \left(1 - \frac{1}{S_{\text{epoch}}}\right)\right). \end{aligned} \tag{9}$$

C. Dynamic Dictionary and Data Augmentation

By constraining z-score z_w with the impact factor, the anomaly probability of the data can change with the iterations. At initialization, all data are assigned the same anomaly probability. As the training progresses, data are assigned different anomaly probabilities, resulting in different anomaly labels. Data labeled 0 are marked as "pseudo-normal data," while data labeled 1 are marked as "pseudo-anomalous data." The "pseudo-normal data" and "pseudo-anomalous data" are expressed as

$$\begin{aligned} D^{d_0} &= \left\{D_1^{d_0}, D_2^{d_0}, \ldots, D_t^{d_0}\right\} \\ D_t^{d_0} &= \left\{x_{d_{0t}}^1, x_{d_{0t}}^2, \ldots, x_{d_{0t}}^K\right\} \\ D^{d_1} &= \left\{D_1^{d_1}, D_2^{d_1}, \ldots, D_k^{d_1}\right\} \\ D_k^{d_1} &= \left\{x_{d_{1k}}^1, x_{d_{1k}}^2, \ldots, x_{d_{1k}}^K\right\} \\ w &= t + k. \end{aligned} \tag{10}$$

During this phase, assigning pseudo-labels to the data can achieve the classification of normal and abnormal data. While "pseudo-normal data" are normal data as much as possible, it cannot be guaranteed that "pseudo-abnormal data" are necessarily abnormal data. This can lead to disastrous consequences for the discriminator $\mathbb{D}_A$, which is specifically trained to detect abnormal data; it is necessary to reextract the abnormal data from the discriminator $\mathbb{D}_A$. Here, a dynamic dictionary D^d is introduced. As mentioned earlier, the error e_w is greater than the mean $M(E)$, and the probability of the data $P(x_w)$ being labeled as anomalous is greater than 0.5. Therefore, the dynamic dictionary is sorted based on the reconstruction error, and data with the top L largest errors are considered as real abnormal data and used for training the discriminator $\mathbb{D}_A$. Then, the dynamic dictionary D^d can be obtained as follows:

$$D^d = \left\{D_1^d, D_2^d, \ldots, D_L^d\right\}. \tag{11}$$

This forces the discriminator $\mathbb{D}_A$ to learn to distinguish real abnormal data, thereby improving its ability to recognize abnormal data.

During each iteration, the discriminator $\mathbb{D}_A$ is updated using L real anomalous data from the dynamic dictionary and reconstructed anomalous data

$$\begin{aligned} D^d &= \text{TOP}_L\left\{D_1^d, D_2^d, \ldots, D_L^d, D_1^{d_{1k}}, D_2^{d_{1k}}, \ldots, D_k^{d_{1k}}\right\} \\ &= \left\{D_1^d, D_2^d, \ldots, D_L^d\right\}. \end{aligned} \tag{12}$$

Through this iterative process, the generator G and the discriminator $\mathbb{D}_A$ engage in a game of optimization, continually

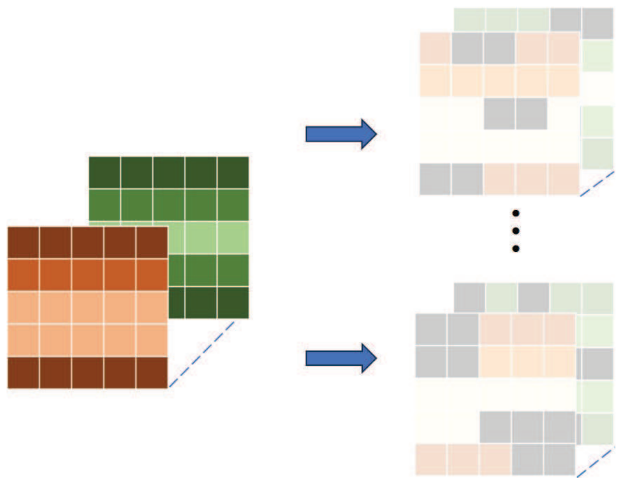

Fig. 4. Data augmentation model. Data stored in the dynamic dictionary (left) and $\widetilde{D}^d$ after data augmentation (right).

improving their performance and achieving the recognition and generation of anomalous data. Eventually, at the end of the training, the generator G will be able to encode anomalous data, and the discriminator $\mathbb{D}_A$ will accurately identify anomalous data and save it in the dynamic dictionary for use in the next training cycle.

Data augmentation can increase the quantity and quality of data in situations where data is scarce. This article adopts a random mask data augmentation method that follows a geometric distribution, as shown in Fig. 4, where black represents the masked state denoted by 0 and color represents the unmasked state denoted by 1. Given the mask ratio r and mask length l_m, $p_{s_t:s_{t+1}}$ represents the noise transition probability from time t to $t+1$, and the complete transition matrix can be expressed as

$$P = \begin{bmatrix} p_{0:0} & p_{0:1} \\ p_{1:0} & p_{1:1} \end{bmatrix} = \begin{bmatrix} 1 - \frac{1}{l_m} & \frac{1}{l_m} \cdot \frac{r}{1-r} \\ \frac{1}{l_m} & 1 - \frac{1}{l_m} \cdot \frac{r}{1-r} \end{bmatrix}. \tag{13}$$

Positive sample pairs are obtained by randomly applying two independent masking operations to each dimension. M is a binary noise mask matrix that follows a geometric distribution:

$$\widetilde{D}^d = D^d \otimes M \tag{14}$$

where $\widetilde{D}^d$ represents the transformed multidimensional time series after data augmentation

$$L_{\text{con}} = -\log \frac{\exp\left(\cos\left(\widetilde{D}^d \cdot \widetilde{D}^{d+}\right)/\varpi\right)}{\sum_{i=0}^{S} \exp\left(\cos\left(\widetilde{D}^d \cdot \widetilde{D}^{di}\right)/\varpi\right)} \tag{15}$$

where $\widetilde{D}^{d+}$ denotes the positive pairs and S represents the total number of samples.

D. Dual-Discriminator

In this section, we propose novel designs for the discriminators $\mathbb{D}_A$ and $\mathbb{D}_N$ with the aim of reducing the distance between normal and reconstructed data and increasing the distance between anomaly and reconstructed data. To achieve this goal, we employ different loss functions for $\mathbb{D}_A$ and $\mathbb{D}_N$. For discriminator $\mathbb{D}_N$, which takes normal data and their corresponding reconstructions as inputs, we design its loss function to gradually reduce the distance between normal and reconstructed data via backpropagation after each iteration of training

$$\begin{aligned} L_N &= \text{BCEloss}\left(D^{d_0}\right) + \text{BCEloss}\left(G\left(D^{d_0}\right)\right) \\ &= -\frac{1}{t}\sum_{i=1}^{t}\left[l_i \log\left(D_i^{d_0}\right) + (1-l_i)\log\left(1-D_i^{d_0}\right)\right] \\ &\quad - \frac{1}{t}\sum_{i=1}^{t}\left[l_i \log\left(G\left(D_i^{d_0}\right)\right) + (1-l_i)\log\left(1-G\left(D_i^{d_0}\right)\right)\right]. \end{aligned} \tag{16}$$

However, the original data are marked as real, and the reconstructed data are marked as fake; then, the original data are labeled as 1 and the reconstructed data are labeled as 0. The loss of discriminator $\mathbb{D}_N$ can be defined as

$$\begin{aligned} L_N &= \text{BCEloss}\left(D^{d_0}\right) + \text{BCEloss}\left(G\left(D^{d_0}\right)\right) \\ &= -\frac{1}{t}\sum_{i=1}^{t}\left[\log\left(D_i^{d_0}\right) + \log\left(1-G\left(D_i^{d_0}\right)\right)\right]. \end{aligned} \tag{17}$$

For discriminator $\mathbb{D}_A$, which takes anomaly data D^d and their corresponding reconstructions as inputs, we design its loss function to further increase the distance between anomaly and reconstructed data after each iteration of training

$$\begin{aligned} L_A &= \text{BCEloss}\left(\widetilde{D}^d\right) - \text{BCEloss}\left(G\left(\widetilde{D}^d\right)\right) \\ &= -\frac{1}{L}\sum_{i=1}^{t}\left[l_i \log\left(\widetilde{D}_i^d\right) + (1-l_i)\log\left(1-\widetilde{D}_i^d\right)\right] \\ &\quad + \frac{1}{L}\sum_{i=1}^{t}\left[l_i \log\left(G\left(\widetilde{D}_i^d\right)\right) + (1-l_i)\log\left(1-G\left(\widetilde{D}_i^d\right)\right)\right]. \end{aligned} \tag{18}$$

Similar to discriminator $\mathbb{D}_N$, the original data are labeled as 1 and the reconstructed data are labeled as 0; the loss of discriminator $\mathbb{D}_A$ can be expressed as

$$\begin{aligned} L_A &= \text{BCEloss}\left(\widetilde{D}^d\right) + \text{BCEloss}\left(\widetilde{D}^d\right)\right) \\ &= -\frac{1}{L}\sum_{i=1}^{t}\left[\log\left(\widetilde{D}_i^d\right) - \log\left(1-G\left(\widetilde{D}_i^d\right)\right)\right]. \end{aligned} \tag{19}$$

Through the training of discriminators $\mathbb{D}_A$ and $\mathbb{D}_N$, we are able to simultaneously reduce the distance between normal and reconstructed data and increase the distance between anomaly and reconstructed data.

E. Adaptive Weighted

This section proposes a novel training function based on reconstruction error for anomaly detection, which requires precise reconstruction of data to retain normal information while discarding anomaly information, to reduce the distance between normal and reconstructed data, and to expand the distance between anomaly and reconstructed data. To achieve this, we redesign the training function to differentiate the contributions of normal and anomaly data to the loss function. The function is formulated as

$$\begin{aligned} L_{\mathrm{re}} &= \frac{1}{w}\left(\sum_{i=1}^{t}\left\|D_i^{d_0}-\hat{D}_i^{d_0}\right\| * \left(1-y_i^{d_0}\right)\right. \\ &\quad \left. - \sum_{i=1}^{k}\left\|D_i^{d_1}-\hat{D}_i^{d_1}\right\| * \left(1-y_i^{d_1}\right)\right) \\ &= \frac{1}{w}\left(\sum_{i=1}^{t}\left\|D_i^{d_0}-\hat{D}_i^{d_0}\right\|\right) \end{aligned} \tag{20}$$

where w is the total amount of all data and y_i is the true label of D_i, meaning that only true data will contribute to the reconstruction loss. However, in an unsupervised learning scenario, labels are unavailable, and it is impossible to separate normal samples from abnormal ones. To solve this problem, we propose a new reconstruction objective specifically designed for the anomaly detection task

$$\delta_i = \frac{e^{-z_i}}{\sum_{a=1}^{w} e^{-z_a}} \tag{21}$$

where the reconstruction error is larger than the average, and the data will receive smaller weights; the reconstruction error is smaller than the average, and the data will receive larger weights. Smaller reconstruction errors correspond to larger weights, and larger reconstruction errors correspond to smaller weights. We also introduce a balancing factor, with the final weight being

$$\delta_i = \frac{e^{-z_i}}{\sum_{a=1}^{w} e^{-z_a}} \cdot \frac{1}{N}\frac{\sum_{a=1}^{w} e^{-z_a} + \left(S_{\text{epoch}}-1\right)\cdot e^{-z_i}}{S_{\text{epoch}}\cdot e^{-z_i}} \tag{22}$$

where N is a normalization factor. At the initial State, the weights and the contribution to the loss function are equal. As the epoch increases, the weights will gradually approach Equation (21). Ultimately, Equation (20) is approximated by

$$L_{\mathrm{re}} = \sum_{i=1}^{w}\left\|D_i-\hat{D}_i\right\|\cdot\delta_i. \tag{23}$$

The reconstruction loss function of Equation (23) allows DDGAN to focus more on fitting the data.

F. Model Training

Given a training dataset $T = \{x_1, x_2, x_3, \ldots, x_T\}$, this article proposes an alternate training approach for a generator and two discriminators, $\mathbb{D}_A$ and $\mathbb{D}_N$, where the training loss functions for discriminators $\mathbb{D}_A$ and $\mathbb{D}_N$ are defined in Equation (19) and Equation (17), respectively. Unlike traditional GANs, where only an adversarial loss function is used to train the Generator, this article utilizes both discriminators $\mathbb{D}_A$ and $\mathbb{D}_N$ and obtains their respective adversarial loss functions. By combining these two adversarial loss functions, a double discriminator-based adversarial loss function is derived, as shown in the following:

$$L_{a\mathbb{D}_N} = -\frac{1}{T}\sum_{i=1}^{T}\log \mathbb{D}_N\left[G\left(x_i\right)\right]$$

$$L_{a\mathbb{D}_A} = \frac{1}{T}\sum_{i=1}^{T}\log \mathbb{D}_A\left[G\left(x_i\right)\right] \tag{24}$$

$$\begin{aligned} L_{ad} &= L_{a\mathbb{D}_N} + L_{a\mathbb{D}_A} \\ &= -\frac{1}{T}\left[\sum_{i=1}^{T}\log \mathbb{D}_N\left[G\left(x_i\right)\right] - \sum_{i=1}^{T}\log \mathbb{D}_A\left[G\left(x_i\right)\right]\right]. \end{aligned} \tag{25}$$

Algorithm 1 Pseudo-Label Generation Algorithm

Require: Sequence data $D = \{D_1, D_2, \ldots, D_N\}$
Ensure: Pseudo-labels $l = \{l_1, l_2, \ldots, l_N\}$
$D \leftarrow G(D)$ ▹Apply graph-based model $G(\cdot)$ for representation or reconstruction
Compute error e_i for each D_i using 4
for each sample D_i **do**
  Compute probability $P(D_i)$ using 9
  Sample $\eta \sim$ Uniform(0, 1)
  if $\eta > P(D_i)$ **then**
    Set pseudo-label $l_i \leftarrow 0$
  else
    Set pseudo-label $l_i \leftarrow 1$
  end if
end for
return l

Algorithm 2 Adversarial Training Algorithm

Require: Fragmented training data $T = \{x_1, x_2, \ldots, x_T\}$, test data $X = \{(x_{1t}, \ldots, x_{it})\}$, max epochs P, adversarial loss rate ν
Ensure: Trained Generator G, Discriminators $\mathbb{D}_A$, $\mathbb{D}_N$
Initialize weights of G, $\mathbb{D}_A$, and $\mathbb{D}_N$
for each epoch from 1 to P **do**
  for each fragment index κ from 1 to n **do**
    $\mathbb{D}_N$ receives D^{d_0} from Equation 10
    Compute loss L_N via Equation 17
    Update $\mathbb{D}_N$ using L_N
    $\mathbb{D}_A$ receives D^d from Equation 12
    Compute loss L_A via Equation 19
    Update $\mathbb{D}_A$ using L_A
  end for
  Compute L_{con} via Equation 15
  Compute L_{re} via Equation 23
  Compute L_{dgan} via Equation 26
  Update Generator G with combined loss
end for
for each test fragment x_{it} in X **do**
  Generate reconstruction: $\hat{x}_{it} = G(x_{it})$
  Compute reconstruction score:
  score = $\sqrt{(x_{it}-\hat{x}_{it})^2}$
end for

However, the double discriminator-based adversarial loss function ignores the temporal correlation of time series data. Therefore, this article combines Equation (23) and Equation (25) to obtain the generator's loss function, as shown in the following:

$$L_{\mathrm{dgan}} = L_{\mathrm{re}} + \vartheta\cdot L_{ad} + \varsigma L_{\mathrm{con}} \tag{26}$$

TABLE I
DATASETS

Dataset	Train	Test	Dim	Anomalies
MSL	58,317	73,729	55	10.72%
SMAP	135,183	427,617	25	13.13%
SWaT	496,800	449,919	51	11.98%
WADI	1,048,571	172,801	123	5.99%
SMD	708,405	708,420	38	4.16%
PSM	105,984	87,841	25	27.8%

where ϑ is used to adjust the weight of L_{ad} and ς is used to adjust the weight with L_{con}. The specific training process is presented in Algorithm 2.

V. EXPERIMENTAL RESULTS

This section conducts extensive experiments to evaluate the effectiveness of DDGAN, with additional experimental results and analyses provided in the Appendixes (see the Supplementary Material).

A. Experimental Setup

1) Public Datasets and Baselines: We evaluate our method on six public datasets: MSL [33] (NASA's Mars rover mission data), SMAP [34] (soil moisture satellite data), SMD [35] (server performance metrics), SWaT [36] (water treatment ICS data), WADI [37] (water distribution network simulation), and PSM [38] (eBay's server metrics). Details of the dataset information are shown in Table I. These datasets cover diverse domains, including space exploration, environmental monitoring, industrial systems, and cloud computing, providing comprehensive benchmarks for anomaly detection.

For comparison, we select seven state-of-the-art unsupervised anomaly detection methods: USAD [39] (adversarial AEs), OmniAnomaly [35] (stochastic RNN), TranAD [40] (transformer-based), GDN [41] (graph neural network), BeatGAN [9] (beat-aware GAN), FGANomaly [10] (pseudo-label enhanced GANomaly), and Anomaly Transformer [42] (self-attention based). These baselines represent major technical directions in time series anomaly detection, including reconstruction-based, GAN-based, and attention-based approaches.

2) Evaluation Metrics: We adopt precision (P), recall (R), $F1$-score ($F1$), and the area under the precision–recall curve (AUC) to evaluate the performance. These metrics are defined as follows: P measures the proportion of true positive samples among all samples, and it is calculated as

$$P = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{27}$$

where TP is the number of true positives and FP is the number of false positives.

TABLE II
AVERAGE PRECISION, RECALL, $F1$-SCORE, AND AUC OF DIFFERENT METHODS ON SIX PUBLIC DATASETS

Method	P	R	F1	AUC
PCA	0.7270	0.6138	0.6536	0.7383
USAD	0.8997	0.8605	0.8736	0.8665
Omni Anomaly	0.7772	0.8377	0.7893	0.8162
TranAD	0.8153	0.9151	0.8418	0.8440
GDN	0.7547	0.8792	0.7860	0.8198
BeatGAN	0.8283	0.8356	0.8263	0.8446
FGANormaly	0.9239	0.8748	0.8885	0.8813
AT	0.9114	0.9241	0.9171	0.9092
DDGAN	**0.9289**	**0.9868**	**0.9564**	**0.9622**

R measures the proportion of true positive samples among all positive samples, and it is calculated as

$$R = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{28}$$

where FN is the number of false negatives.

$F1$ is the harmonic mean of P and R, and it is calculated as

$$F1 = 2 * \frac{P * R}{P + R}. \tag{29}$$

B. Tasking Setting

In experiments, the sliding window width is uniformly set to 150. The model is trained using the Adam optimizer with a learning rate of $1\mathrm{e}^{-4}$. The RNN encoder has an RNN hidden dim of 50, and the decoder consists of multiple linear layers with a hidden dim of 50. The hyperparameter is set to 0.5. The training was conducted in a Windows environment equipped with a 2.50-GHz 12th Gen Intel[1] Core[2] i5-12400F CPU and an 8-GB NVIDIA GeForce RTX 3060 GPU. The training dataset is divided into a training set and a validation set in a 7:3 ratio to obtain the optimal model and avoid overfitting or underfitting. As the model is considered an unsupervised learning model, it cannot directly adjust hyperparameters based on the $F1$-score on the validation set as in supervised learning. To avoid reliance on predetermined thresholds, we can search for all possible thresholds to determine the normality and abnormality of observed values and select the optimal $F1$-score as the performance metric for model evaluation. We obtain the anomaly scores in the test set and evaluate them individually as thresholds. Finally, the threshold yielding the best $F1$-score is selected as the final result. Then, we conducted multiple experiments based on the final selected model and used the mean and variance of the multiple experiments as the final results of the model.

C. Overview Performance

We conducted an extensive evaluation of DDGAN in six datasets, comparing its performance with multiple baseline

[1]Registered trademark.
[2]Trademarked.

TABLE III

PERFORMANCE OF EIGHT UNSUPERVISED ANOMALY DETECTION METHODS COMPARED TO DDGAN ACROSS SIX PUBLIC DATASETS. IT SHOWS PRECISION, RECALL, $F1$-SCORE, AND AUC FOR EACH METHOD, WITH THE BEST RESULTS HIGHLIGHTED IN BOLD

Dataset	**MSL**				**PSM**				**SWAT**			
	P	**R**	**F1**	**AUC**	**P**	**R**	**F1**	**AUC**	**P**	**R**	**F1**	**AUC**
PCA	**0.9361**	0.8401	0.8855	0.8774	0.7849	0.9925	0.8766	0.8684	0.3865	0.3288	0.3553	0.5675
USAD	0.8943	0.8764	0.8853	0.8946	0.9062	**0.9979**	0.9498	0.8974	**0.9973**	0.6871	0.8136	0.8292
Omni Anomaly	0.7845	**0.9921**	0.8762	0.8751	0.8839	0.7447	0.8084	0.7983	0.9782	0.6952	0.8128	0.8228
TranAD	0.9034	0.9781	**0.9393**	0.8601	0.9265	0.9978	0.9608	0.9288	0.9972	0.6879	0.8142	0.7952
GDN	0.9002	0.8995	0.8998	0.8789	0.9018	0.9362	0.9187	0.9024	0.9698	0.6595	0.7851	0.7977
BeatGAN	0.8913	0.8633	0.8771	0.8621	0.9035	0.9384	0.9206	0.8772	0.7364	0.9291	0.8216	0.8813
FGANormaly	0.9104	0.8985	0.9044	0.8791	0.9492	0.9873	0.9679	0.9187	0.9850	0.7952	0.8800	0.8666
AT	0.9126	0.8585	0.8847	0.8603	0.9039	0.9382	0.9207	0.9257	0.9146	0.9522	0.9330	0.8958
DDGAN	0.8817	0.9516	0.9153	**0.9192**	**0.9889**	0.9874	**0.9882**	**0.9888**	0.8906	**0.9826**	**0.9343**	**0.9369**
Dataset	**SMAP**				**WADI**				**SMD**			
	P	**R**	**F1**	**AUC**	**P**	**R**	**F1**	**AUC**	**P**	**R**	**F1**	**AUC**
PCA	0.9092	0.5760	0.7052	0.7554	0.5668	0.4956	0.5288	0.6564	0.7787	0.4498	0.5702	0.7046
USAD	0.8390	0.8800	0.8590	0.8449	0.8552	0.7241	0.7842	0.7938	0.9063	0.9977	0.9498	0.9391
Omni Anomaly	0.8131	0.9414	0.8726	0.8797	0.3152	0.6544	0.4255	0.5496	0.8885	0.9986	0.9403	0.9718
TranAD	0.8044	**0.9999**	0.8916	0.9025	0.3526	0.8297	0.4949	0.6076	0.9074	0.9972	0.9502	0.9699
GDN	0.7482	0.9899	0.8522	0.8893	0.2910	0.7930	0.4258	0.5207	0.7171	0.9972	0.8343	0.9300
BeatGAN	0.7890	0.9363	0.8564	0.8724	0.7286	0.6169	0.6681	0.7530	0.9209	0.7298	0.8143	0.8217
FGANormaly	0.7601	0.9995	0.8635	0.9000	**0.9639**	0.6163	0.7519	0.7679	0.9747	0.9519	0.9632	0.9556
AT	**0.9259**	0.9751	**0.9499**	0.9401	0.9170	0.8663	0.8909	0.8790	0.8946	0.9542	0.9234	0.9545
DDGAN	0.8818	0.9993	0.9369	**0.9629**	0.9306	**1.0000**	**0.9640**	**0.9653**	**0.9996**	**1.0000**	**0.9998**	**0.9998**

TABLE IV

PRECISION, RECALL, AND $F1$-SCORES PERFORMANCE OF LSTM-AE, LSTM-$\mathbb{D}_N$, LSTM-$\mathbb{D}_A$, DDGAN, DDGAN, AND DDGAN IN FOUR PUBLIC DATASETS

DATASET	MSL			SMAP			SWaT			SMD		
METRICX	P	R	F1	P	R	F1	P	R	F1	P	R	F1
LSTM-AE	92.80	78.88	85.28	74.51	95.43	83.69	73.73	25.39	37.77	38.39	39.35	38.86
LSTM-$\mathbb{D}_N$	91.39	88.41	89.88	76.59	**99.99**	86.75	**91.78**	92.50	92.14	56.83	66.89	61.45
LSTM-$\mathbb{D}_A$	84.39	79.02	81.62	80.84	92.06	86.08	86.42	85.24	85.82	38.51	38.83	38.67
ddGAN	92.01	85.42	88.60	80.80	99.64	89.23	89.00	96.81	92.74	50.39	79.66	61.73
DdGAN	**92.91**	88.83	90.83	86.70	94.64	90.50	87.07	**98.26**	92.32	99.92	99.90	99.91
DDGAN	88.17	**95.16**	**91.53**	**88.18**	99.93	**93.69**	89.06	**98.26**	**93.43**	**99.96**	**100.00**	**99.98**

methods. As shown in Table II, DDGAN significantly outperformed eight other unsupervised anomaly detection methods across these six public datasets, demonstrating its exceptional performance in unsupervised anomaly detection tasks. Table III also provides a detailed comparison of DDGAN and other baseline models on public datasets. In the MSL dataset, our method achieved an AUC score of 91.92%, surpassing all baselines by an improvement of 2.46%. Although the $F1$-score was not the highest, it still exhibited suboptimal performance. In the PSM dataset, DDGAN excelled in accuracy, $F1$, and AUC metrics, with an impressive increase of 2.03% in the $F1$-score over the baseline, achieving an AUC of 98.88%, a 6% improvement. Despite a slightly lower recall compared to USAD, our recall rate remained high at 98.51%. On the SWAT dataset, DDGAN achieved the best results across recall, $F1$-score, and AUC, with a recall improvement of 3.04% over the baseline. For the WADI dataset, our method also demonstrated outstanding performance in recall, $F1$-score, and AUC, successfully achieving a perfect recall rate and a 7.31% increase in $F1$-score compared to baseline. In the SMD dataset, DDGAN exhibited excellent performance across all evaluation metrics, achieving an accuracy of 99.96%, a recall rate of 100.00%, an $F1$-score of 99.97%, and an AUC of 99.97%, highlighting its significant advantages over other methods. However, in the SMAP dataset, our method performed poorly. This was primarily due to the emphasis on temporal relationships while failing to adequately consider intersensor correlations, which resulted in unsatisfactory accuracy and recall rates. Nonetheless, the AUC metric still reached optimal levels, while the $F1$-score achieved suboptimal results. In summary, our method demonstrated superior performance across most datasets, outperforming several baseline methods across various evaluation metrics, underscoring the practicality and effectiveness of unsupervised anomaly detection tasks. It is crucial to note that the pseudo-label generation in this study relies on a reconstruction error-based dynamic dictionary screening mechanism, theoretically grounded in the core hypothesis that "normal samples exhibit significantly lower reconstruction errors than anomalies." However, this mechanism shows high sensitivity to data distributions: in scenarios with severe class imbalance, fixed-capacity dictionaries easily lead to insufficient representation of normal samples, thereby generating pseudo-label noise. Additionally, the mean squared error (mse) used for reconstruction error calculation has inherent limitations, resulting in inadequate sensitivity to localized subtle anomalies.

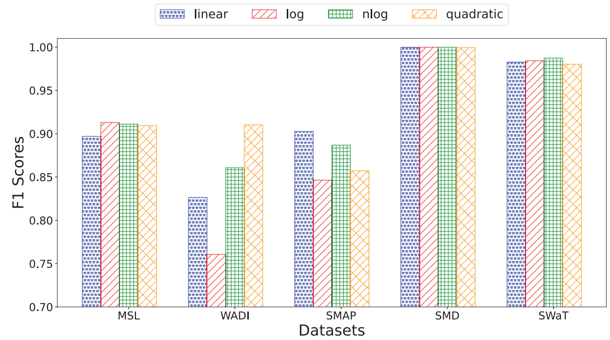

Fig. 5. Mean values achieved by different methods under various parameter configurations.

D. Ablation Study

DDGAN demonstrates significant improvements over most baseline models. This section presents ablation experiments designed to validate the effectiveness of various modules, structures, and methods within the algorithm. Specifically, we compared DDGAN with five other models: LSTM-AE, LSTM-$\mathbb{D}_N$, LSTM-$\mathbb{D}_A$, and ddGAN. LSTM-AE represents a conventional LSTM-based AE, while LSTM-$\mathbb{D}_N$ integrates a discriminator trained on normal data into the LSTM-based AE and LSTM-$\mathbb{D}_A$ employs a discriminator trained on anomalous data. The ddGAN model combines an LSTM-based AE trained exclusively on anomalous data but excludes the dynamic dictionary. The distinction between DdGAN and DDGAN lies in the absence of data augmentation in the dynamic dictionary component. The ablation experiments were conducted on four public datasets: MSL, SMAP, SWaT, and SMD. Table IV presents the results of the ablation experiments. It is evident that DDGAN outperforms LSTM-AE, LSTM-$\mathbb{D}_N$, LSTM-$\mathbb{D}_A$, ddGAN, and DdGAN across the four public datasets, particularly excelling in recall and $F1$-score, indicating that DDGAN possesses stronger generalization capabilities in unsupervised anomaly detection tasks. Furthermore, the results in Table IV indicate that both single-discriminator (handling normal or anomalous data) and dual-discriminator architectures contribute to enhancing anomaly detection performance. Notably, the discriminators for normal and anomalous data show significant improvements across all datasets when compared to the traditional LSTM-AE, highlighting the effectiveness of integrating LSTM-AE with GANs for anomaly detection tasks. The results from ddGAN and DdGAN demonstrate the benefits of incorporating a dynamic dictionary for storing anomalous data to train the anomaly discriminator. DDGAN's inclusion of data augmentation within the dynamic dictionary addresses the challenge of anomalous data scarcity, and the results suggest that this approach is effective in improving model performance. The aforementioned ablation experiments reveal that the combination of the normal data discriminator and the anomalous data discriminator, along with components such as the dynamic dictionary and data augmentation, enhances anomaly detection performance. The dynamic dictionary serves as an auxiliary component for storing limited anomalous data, playing a crucial role during the training process. Meanwhile, the data augmentation component alleviates the issue of data imbalance by enhancing the anomalous data, thereby addressing the scarcity of such data. Overall, the ablation study results indicate that improvements to the model from different perspectives effectively enhance anomaly detection capabilities and performance, and these enhancements are not mutually exclusive.

E. Parameter Sensitivity

This section investigates the impact of selecting different S_{epoch} strategies on the performance of the model. We explore the choices of S_{epoch} from a set of [log, N, $n\,$log, n^2]. We examine various S_{epoch} strategies and present their effects on the model's mean values in Fig. 5. It is observed that, in most cases, employing n^2 strategies leads to improved performance. All four strategies exhibit a monotonically increasing trend, albeit with varying rates of increase as the number of epochs progresses. Strategy n^2 notably demonstrates the highest increase in both optimal and mean values. These findings underscore the appropriateness of the research approach based on reconstruction error. By adopting strategy n^2 for anomaly detection, the model effectively accentuates the discrepancy between abnormal and reconstructed data, thereby yielding a substantial enhancement in the effectiveness of anomaly detection through reconstruction error analysis.

VI. Conclusion

This study addresses the data dependency and overfitting challenges of GANs in multivariate time series anomaly detection by proposing a dynamic dictionary-based GAN (DDGAN) framework. The methodological innovation lies in the synergistic integration of a pseudo-label filtering mechanism and a dynamic dictionary optimization strategy, which collaboratively establishes a representation system for normal and abnormal samples. Specifically, the dynamic dictionary enables adaptive storage of anomalous samples based on reconstruction error thresholds, while the pseudo-label mechanism iteratively refines the screening accuracy of normal data. Notably, the method may encounter challenges in scenarios with class imbalance due to dictionary capacity constraints that could compromise normal pattern characterization. Compared to existing approaches, the core breakthroughs include the dual-discriminator architecture design and the joint optimization mechanism of dynamic weight loss functions, which collectively enhance both normal data reconstruction fidelity and anomaly discrimination capability. Future research will prioritize addressing technical bottlenecks in multimodal feature fusion and fine-grained anomaly classification while exploring adaptive dictionary capacity adjustment strategies to improve model robustness.

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